Statistics 266 – Practice Midterm 1A – Mike Kowalski

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Instructions: This is a closed book exam. Only the formula sheet provided and a calculator are allowed in the exam.

There are 6 questions. All questions have a value of 5 marks. This isn't necessarily a reflection of the difficulty of the question. In order to obtain full credit for any part you must show all your work. Be as clear and concise as possible! **The exam is out of 30.**

NOTE: When asked to find a density you must also specify the support where ***f* > 0**. Feel free to do this by stating the appropriate relations or by drawing a picture.

Question 1: The time (in hours) it takes workers to complete a specialized task is a random variable, Y, described by the following density

$$f(y) = \begin{cases} \frac{3}{2}y^2 + y \ , & 0 < y < 1 \\ 0 \ , & else \end{cases}$$

A financial bonus (in hundreds of dollars) is given out by the following formula, $U = 1 - (Y/2)$. What is the probability density of U?

Question 2: When a current I (in amperes) flows through a resistance R (in ohms), the power generated is given by $W = I^2 R$ (in watts). Suppose that I follows a $Beta(\alpha = 2, \beta = 2)$ distribution and R follows a $Beta(\alpha = 2, \beta = 1)$ distribution. If I and R are independent, determine the density of W.

Question 3: Consider two independent $Exponential(\theta)$ random variables. What is $P(4Y_{(1)} > Y_{(2)})$?

Question 4 : Let X and Y denote the time (in hours) until 2 different batteries fail. Suppose the joint density of X and Y is

$$f(x,y)=\begin{cases}\dfrac{2}{x^2y^2} & ,1<x<y\\ 0 & ,\text{otherwise}\end{cases}$$

$U=Y-X$ and $V=Y/X$ are both measures for how much longer battery Y lasts compared to battery X. (U is the additive measure, V is the multiplicative measure). Find the joint probability density of U and V.

Question 5 : The joint distribution of amount of pollutant emitted from a smokestack without a cleaning device, X, and a similar smokestack with a cleaning device, Y, is given to be

$$f(x,y)=\begin{cases}1, & 0<x<2,\ 0<y<1,\ 2y<x\\ 0, & else\end{cases}$$

Find the density of $U=X-Y$.

Question 6 : A light system has two bulbs in joint operation. Let X and Y denote the lengths of life of the two bulbs. The joint density function for X and Y is

$$f(x,y)=\begin{cases}ye^{-y(x+1)} & ,\ x>0,y>0\\ 0 & ,\ else\end{cases}$$

Consider the random variable $U=XY$. Find the moment-generating-function of U. What is the probability density for U?

3) $P(4\ Y_{(1)} > Y_{(2)})$

$Y_{(1)} =$